

University of Glasgow | School of Computing Science

Level 3 Project Case Study Dissertation

CENSIS Augmented Reality Application

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Abstract

This paper is a case study that explores the software development process of the augmented reality application created for a software company called Censis. This application was created as part of a team project in the 3rd year at the University of Glasgow.

The study provides an overview of the project, detailing the features that were implemented in the augmented reality application created. Such features that will be discussed is the augmented reality overlay, navigation of such app and additional features implemented. The study also provides An insight into the motivation behind the project creation as well as details about the type of customer we were working with to create such a project. The development process of the app is examined and analysed in this study, highlighting the software practices employed during such project. These practices include Agile methodologies, testing strategies, the use of continuous integration/continuous development and other various practices. Such will be explained and discussed in this study. This study also reflects on the challenges faced and solutions devised throughout the development process as well as a reflective analysis of any events that occurred during the project development.

The study will conclude with key achievements gained , the various knowledge gained throughout the project and how such project contributed to the understanding of software development practices.

Education Use Consent

We hereby give our permission for this project to be shown to other University of Glasgow students and to be distributed in an electronic format.

1 Introduction

This paper presents a case study of the development of an augmented reality application that offers an innovative way to interact with historical statues and monuments. The application utilizes location tracking technology to determine the user's proximity to specific statues. When a user is near a designated statue, they are granted access to view a detailed 3D model of the statue in an augmented reality fashion, enhancing their historical and cultural experience.

The application also provides a variety of additional features to enhance the user experience. Users can access real-time weather data specific to Glasgow, enabling them to plan their visits accordingly. Additionally, the application offers login and registration functionalities, allowing users to create personalized accounts for a tailored experience. Furthermore, users have the option to view their recently visited locations and explore the 3D models in a customised augmented reality view.

This document will begin by introducing the type of customer for whom this project was developed, delving into the unique characteristics of the customer. Following this, the paper will detail the final product and its functionalities, providing a comprehensive understanding of the product. Subsequently, this paper will offer a thorough analysis and overview of the software development process, highlighting the software practices employed throughout, such as testing strategies and other practices that contributed to the successful development of the project. Finally the paper will conclude the software development process with a full analysis on the achievements as well as an insight into potential areas of improvement in future projects. Additionally, the conclusion will provide an overview of the overall project experience, comparing it with existing literature or similar projects for a broader perspective.

The case study structure is as follows. Section 2 explains the background of the case study which includes a descriptive analysis of the client as well as the project context, this will include the customer aims and objectives. In addition a overview of the project will also be given which includes the project details and the requirements that were achieved in such a project. Section 3 explores the software development process throughout the project and will reflect on each of the main events that occurred throughout such project. The subsections 3.1 to 3.2 provides an overview of the requirements as well as communication with the customer. Subsequent subsections from 3.3 to 3.4 discuss the team management at various stages, the encountered problems, and the strategies employed for resolution. The team management aspect will also discuss how each issue is distributed and many more factors relating to such. Each subsection reflects on the actions taken, drawing comparisons with existing literature. Finally section 4 will conclude the case study showing the key achievements gained as well as improvements for the future.

2 Case Study Background

2.1 Our Client

CENSIS is Scotland's Innovation Centre for sensing, imaging and Internet of Things (IoT) technologies. They work with private and public organisations of all sizes to de-risk and accelerate innovation and overcome technology barriers to achieve business transformation. They provide professional advice, with the expertises to guide organisations through the complexities of sensor and IoT. They also extend technical support, assisting businesses in navigating the practical challenges of implementing these technologies. A example of this is a development of Pine weevil camera trap system, an IoT device that enabled remote monitoring system to tackle the perennial risk posed to forestry from pine weevils. The customers we worked with on this project were Helen Sandison.

2.2 Project Proposal

Censis, a company focused on bridging the gap between the real world and the digital realm, has tasked us to create an augmented reality (AR) app. This app will empower users to seamlessly interact with both physical and digital environments, utilizing real-time data to enhance their experiences. Censis specifically desires an app that offers toggling between map and camera modes for versatile exploration. Additionally, the app will provide users with detailed information about augmented reality objects or locations, ensuring a comprehensive and informative interaction between the user and the augmented world. Censis also requested other features such as ensuring that users can customise their augmented reality experience. In addition Censis also requested to ensure that we incorporate simulations for specific scenarios. For example, if the weather data says that it is currently raining then the app could present a rainy background for such case.

The final application created will be presented to Censis for display to their customers. Additionally, Censis aims to use this application as both inspiration and a framework for future development of augmented reality applications.

Censis requested for us to research the technology to be used at the start of the project, they presented to us few technologies such as OpenCV[11] as a suggestion. However such technology was not as useful to use in our finalised project. This is because we found from the article that OpenCV was not the scope of this project as such technology is only used for image recognition in this scenario. Censis did not request the camera to be used as it is not accurate and requested map detection as a means to detect if a user can get the augmented reality of a statue or location. Thus we found out Swift UI and React native are 2 very useful technologies for the progression of this project as they both support augmented reality. In particular, our decision to utilize Swift UI was influenced by our research on the 'ReforestAR' application[9], which demonstrated the effectiveness of Swift UI in similar projects.

2.3 The application

The project was created in a form of an mobile application. The reason to such was because many augmented reality applications currently have been implemented on mobile and we found that doing such would be more beneficial and have much more of a correlation to the initial requirements laid down by the customer.

The project first has a login and register page. The user can either login with their account or create a new account to login. Once the user has logged in, the users are displayed with a menu page of the models they have recently viewed as well as real time weather data. This real time weather data is a part of the requirements the customer has asked and thus the reason why such was implemented. In addition the main page background changes in accordance to the weather data, this is to create a unique simulation for specific scenarios and thus completes a requested section from the customer. An image of this will be shown in figure 1.

The user can then access multiple tabs at the bottom of the screen. One of these tabs include a navigation link to the map page where users will be allowed to navigate to one of the pin annotations displayed on the map. These pin annotations each represent a specific historical or famous location in Glasgow. Users can navigate to such location and get a augmented reality view of models representing such locations. An image of this will be shown in figure 2. Users can also view historical information of such models as well as interact with these models. An image of this will be shown in figure 3. These models can be interacted via turning or zooming into these models.

The functionality above covers majority of the requested requirements from the customer. The ability to interact with the 3d model is a requirement and such have been covered as explained above. In addition map navigation is also a requirement the customer has requested which has been covered above. These are the main requirements. Other small requested ideas were covered such as text annotations which allows the user to view historical descriptions of a 3d model which has been developed as mentioned above as well as customizable scenarios. Scenarios can be customised if user selects a color of their choice, this will change the background of the augmented reality view to be the selected color and the users can view the augmented reality with such color. An image of this will be shown in figure 4.

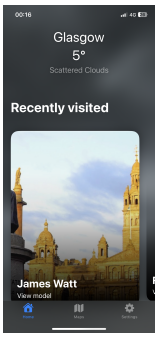


Figure 1: Main Menu



Figure 2: Augmented Reality

The project requirements and the final application were inspired by existing works

and research in the field. One notable example is a mobile app designed for cultural heritage reactivation[6], where augmented reality and location tracking were integrated functionalities. This prior work demonstrates similarities in functionality and purpose with our application. Our aim was to create an app that enhances learning experiences, particularly targeting students. This aligns closely with the objectives of the aforementioned app, showcasing a shared vision in leveraging technology to enrich educational experiences.



Figure 3: 3d Model Description



Figure 4: Customizable Background

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3 Reflections

3.1 Requirements and Features

After our initial customer meeting, we recognized that our project proposal lacked specificity and allow various approach. Over the course of two weeks of stand-up meetings, we refined our project idea and secured confirmation from the customer. We create user stories with a focus on enhancing user experience and outlining the program's features for the customer's approval. Additionally, we developed a Figma prototype to show what customer can expect for the program's user interface 1 week after the requirement confirmation. We are confident that this systematic approach has laid a solid foundation for our project requirements and features.

After establishing a solid foundation, our next step involved researching the technologies required to develop the outlined project, particularly focusing on mobile application development and AR integration, areas in which none of our team members had prior experience. Initially, we opted to use React Native for mobile app development. However, as we delved deeper into the integration of augmented reality (AR), we encountered challenges, notably the inability to use ARKit. These difficulties will be further discussed in the Technical Issues section.

After conducting thorough research on the technologies to be employed, we proceeded to prioritize the features in alignment with the customer's requirements. Utilizing the MOSCOW method[5], which categorized each feature into Must have, Should have, Could have, and Would like to have. This approach enabled us to focus on

developing the essential features necessary for the minimum viable product (MVP) [7], ensuring a effective development and viable product.

After discussing with the customer, we reached a decision to place object recognition in the 'Would like to have' category. This decision was based on the realization that object recognition may exceed our current capabilities and time constraints. Instead, we focus on implementing geolocation-based features, which are more achievable within the project timeline. This experience taught us the importance of realistic scope management and prioritization, helping us to focus on what could be accomplished within our constraints.

Throughout the software development process, we maintained communication with the customer, consistently seeking clarification on each feature or constraint of the application. By actively engaging with customer, we ensured a thorough understanding of the customer's requirements and expectations. This collaborative approach not only provided transparency but also allowed us to address any potential issues or changes in a timely manner.

Through discussions with the customer, we explored the incorporating simulations for specific scenarios. This led us to consider integrating a real-time data API which has numerous options. With ongoing communication and agreements with the customer, we decided to utilize the weather API to enhance the functionality of our application.

Upon reflecting on our progress, we have gained a lot of valuable lesson. We learned that it's important to be clear about what we're building from the start. When we researched the technology we needed, we found out that some things were harder than we thought, like adding augmented reality (AR). We also learned that we need to be flexible and ready to change our plans if something doesn't work. By deciding which features were most important for our application to work, we learned how to prioritize and focus on what matters most using MOSCOW method. When we talked with the customer, we learned that it's okay to say no to some things if they're too hard or take too much time. For example, we decided to put object recognition in the 'Would like to have' category because it was too difficult for us right now. Instead, we worked on adding features based on location, which was easier. We learned that communication with the customer is really important. By talking a lot with them, we made sure we understood what they wanted and could make changes quickly if needed.

3.2 Client Communication

Misunderstood requirements for a project are often the main reason why software development is delayed or unreliable, so it's vital to communicate efficiently with the client to ensure that the project's requirements are correctly understood [4].

3.2.1 Communication Strategy

In order to ensure that the project development is aligned with the requirements of client as well as catering for potential changes in the client's requirements, the communication model with the client we have adopted the Agile development approach - Scrum[12]. The main aspects as follow:

- **Sprint Planning and Reviews**. We set up the development cycle in three-week sprints, allowing us to break the project down into manageable parts. At the beginning of each sprint, we held planning meetings with CENSIS representatives to discuss upcoming tasks and reprioritise based on feedback. Sprint reviews at the end of each phase provided an opportunity to showcase new features and functionality, ensuring that the project was aligned with the client's expectations.
- **Regular Stand-ups**. Weekly stand-up meetings within our team help keep everyone informed of progress and any emerging issues. Whilst these are internal, they ensure that we are able to provide updates to CENSIS in a timely and consistent manner, thus promoting open lines of communication.
- **Meetings**. In addition to the retrospective meetings held at the end of each sprint, we usually have face-to-face sessions(Table 1) in the week leading up to the sprint retrospective meeting to give users a first-hand look at the current state of the system, as well as suggesting changes to ensure that the customer stays engaged in the project development.
- **Meeting Records**. There was a clear division of roles for the members of the meetings, and to reduce the cost of familiarisation we used the same roles for each meeting. At the end of each meeting, a note taker (Wenjie Yang) uploaded the minutes to the GitLab wiki.

Data	**Meeting Title**	**Form**
04/10/2023	Project Requirements	Zoom
20/11/2023	CENSIS Introduction & Requirements Confirmation	In Person (CENSIS office)
29/11/2023	UI Design and Figma Prototype	Zoom
17/01/2024	Real Time Map & Interaction	Zoom
07/02/2024	Customer Application Interaction	In Person (Lab 720)
14/02/2024	Customize Scenario and Finish the Project	Zoom
19/03/2024	Project Handover Discussion	Teams
20/03/2024	Final Demonstration Presentation	In Person

Table 1: Structured Meetings

3.2.2 Communications Issue and Solutions

Although the communication strategy used improved the communication strategy to a certain extent, it still did not fully address the deviation in the understanding of

the requirements due to the client's brief description of the requirements and the fact that our targeted questions did not cover all the content.

In the third Zoom meeting (14/02/2024) we implemented the use of geolocation to recognise buildings, however this was passive, i.e. only when the user was ready to access the 3D model of the building in question, and lacked the proactive recognition that the client expected, e.g. proactively prompting the user when they passed by the building, e.g. through notifications, etc. This made it necessary to postpone the recognition of the 3D model to a later date. This caused us to delay 3D model recognition until the last sprint, however, at the same time we also needed to switch the front-end technology stack from React Native to Swift UI for better iOS support (dropping Android). This required us to spend additional time to understand how to implement iOS message notifications and the mechanism of running the app in the background, however, the remaining time was not enough to justify spending more time on this, so we discussed with the client to remove the sub-functionality and keep the passive recognition. Fortunately, the client accepted our proposal and supported us in switching technology stacks.

3.2.3 Reflective Thought

Reflecting on the communication challenges encountered during the project, it became clear that despite our structured communication strategy, there were still gaps in understanding and aligning with client expectations. The events surrounding the implementation of the geolocation functionality highlighted a serious oversight in our initial requirements gathering process. Whilst we had targeted questions, we failed to fully reflect the client's vision, in particular their expectations for proactively building architectural identity. This oversight required us to make significant adjustments (tech stack tweaks) along with the associated feature implementations, increasing our workload as well as exceeding our capacity.

This experience emphasised the importance of not only asking the right questions, but also ensuring that there is a common understanding of the project requirements between the client and the development team. It taught us the value of flexibility and the need to be prepared to adapt to new insights and requirements. In addition, it emphasises the importance of clear, comprehensive targeted questions and the need for regular, in-depth discussions with the client to clarify and refine project objectives.

This reflective process will help shape our approach to client communications and project management. It reinforces the need for a more dynamic and responsive communication strategy to adapt to the changing nature of software development projects. Drawing on the lessons learnt from this project, we are now in a better position to anticipate and resolve potential misunderstandings early in the development process, ensuring that projects are completed in a more consistent and efficient manner.

3.3 Team management

Our projects use agile methods, specifically version control methods, to improve workflow efficiency and team collaboration. Our team uses a distributed version control system (Git), and when we need to develop a new feature or fulfil a new request from a customer, we first create a new issue in gitlab. When assigning tasks, each member is responsible for one branch, and each branch corresponds to one issue. After completing the assigned task, the member uploads the code to the local branch, initiates a merge request to the master branch, which is then reviewed by team members, and finally merges the code into the branch. Once the merge request is approved and the merge is successful, the member can close the completed issue. This event-driven branching model does its best to avoid code conflicts and allows multiple team members to work simultaneously.

In addition to constantly creating new issues and solving new issues, summarising and reviewing is also a very important part of our group's tasks.

After each client meeting, the meeting note taker collates the content of the meeting, and then we have a retrospective meeting where we prioritise the events based on the meeting notes and add them to the gitlab issue board. There are 8 boards: Open, Priority, Subtask, Backlog, TODO, In Progress, Completed, and Closed. In the Summary, we will prioritise the current backlog tasks and assign them to the TODO, In Progress, Completed and Closed tabs based on their status. In the meeting notes, we will give each member 5-10 minutes to describe each member's individual progress, including but not limited to personal difficulties and features that need to be completed in collaboration with other team members. We will summarise these issues in the meeting and discuss solutions on a case-by-case basis to ensure that each member is making consistent progress and completing as much of the backlog as possible.

However, we find ourselves struggling to cope with the diverse backgrounds, skill levels and availability of our members. Whilst this diversity is a strength, it also presents a huge challenge in ensuring effective collaboration and fair distribution of workload. Liang et al. shows that Task conflict is positively connected with team performance and might arise from a diversity of knowledge within the team[8]. And we also encountered situations where members of the group went missing, adding to the difficulty of the work.

In response to diverse backgrounds, skill levels and availability of our members. We divided up the duties and responsibilities among the team members according to their respective skill sets and learning goals. This methodology not only maximises project results but also fosters individual growth and contentment. And also we have tried to be flexible with meetings and deadlines as much as possible, taking into account the different schedules of team members. Members' work outside of the project can be accommodated with this flexibility, which lowers stress and boosts involvement. According to Acuña et al. report that soft factors affecting software development team performance include team climate[1]. Therefore, in a team project we need to make sure that each member works in a good group environment.Minimise the

creation of internal conflicts that can lead to reduced productivity.

To solve the team member disappeared, our group first discussed this team member's the possibility of the team member returning and completing the task, while redistributing tasks within the team and adding contingency plans to allow team members to communicate about their respective tasks, current developments, and any problems encountered. In this way, we can ensure that in the event of a key member's incapacitation, at least one other member of the team will be able to take on the important responsibilities of the position. Asproni concluded from his findings and said that effective teamwork requires a results-oriented team structure that includes clear team roles and responsibilities, effective team communication, continuous monitoring of member performance, and sound, fact-based judgement when problems are identified[2]. Taking into account of what Asproni concluded from his research, this made us realise that team structure is extremely crucile in terms of creating a viable communication between team members. Hence why we assigned roles and tasks to each specific team members and as well as monitored each team member progress on such tasks to ensure everything is lining up smoothly. This ensured that every team members task would be completed on time and if such is not the case then another team member will provide assistance to ensure the task is completed. This then ensured problems were identified early and enhanced team communication and work speed.

Overall, team management is a complex task, and while a retrospective meeting can help us better manage our team to receive enough feedback to be effective, a retrospective alone is not enough to manage the whole team well. Every step of the way, from our event-driven branching model to dealing with the challenges of team diversity, has provided us with valuable lessons. In managing project tasks and addressing individual and collective challenges through GitLab, we've learned the importance of clear communication. Assigning tasks based on skill sets and learning objectives not only optimised project outcomes, but also fostered personal growth and team cohesion.

In addition, we recognise the importance of shared responsibility and the need to be prepared for contingencies in the event of problems with absent team members. This reduces disruption, ensures project continuity and highlights the value of co-operation and support amongst team members. Our ongoing assessments and updates became the backbone of proactive problem solving and the development of continuous improvement.

3.4 Technical Issues

The augmented reality application developed by our team at the University of Glasgow posed a range of technical challenges that significantly affected project progress and learning outcomes. These challenges, from performance issues to platform-specific support issues, not only tested our problem-solving skills but also informed

our understanding of software development in augmented reality and cross-platform compatibility.

1. **React Native AR iOS support issue and Map crashing on Android**

   At the beginning of our project we used React Native and Expo for cross-platform development, with the goal of streamlining the development process and ensuring consistent application behavior across iOS and Android platforms. We faced major challenge, which is about AR support in React Native Expo for iOS and map feature crashes on Android

   The AR functionality that is important to our application is not well supported in the React Native framework, especially when Expo is installed. This limitation became apparent when integrating AR models with advanced interactions, which was fundamental to the user experience of our application. Together with Expo, the realization that React Native couldn't deliver the AR integration and functionality needed on iOS devices led us to rethink our technology stack

   These platform-specific issues, complicated by the fact that they tend to fall on the side of the map on Android devices, forced a major shift in our development strategy. We made the difficult decision to abandon the cross-platform strategy and switch to native development, since the Swift UI for iOS, Using ARKit's strong support for AR functionality[13]. This change required us to discontinue enhancements achieved through React Native and Expo, including homepage enhancements, details page, login/registration functionality, and other efficiencies

2. **Image Loading Performance** One of the main problems we encountered was the long load times for images downloaded from the cloud. This issue was important because it directly affected the user experience, the image load time was more than 15 seconds. The main cause of this problem was our original design decision to dynamically import images from cloud storage for each instance needed in the application. This approach, while straightforward, did not take into account inherent latency in web requests, or changes in web speed among end users extended loading time is a major issue, especially in an augmented reality context for an application that relies heavily on visual elements to optimize user engagement historical imagery and monuments, where such delays hinder the immersive experience we aimed for to give to users.

   To solve this, we implemented a caching solution. By storing images in an application's cache after an initial load and increasing the cache size, we make sure that these images do not require network fetch for later access, thus reducing load times significantly[14]. Therefore, this product not only improved the performance of the application but also improved the application's performance. It also served as a valuable lesson on how to optimize resource usage in mobile applications, balancing local storage constraints and speed.

3. **3D Models and ARKit Compatibility** A key feature of our application is

the augmented reality visualization of historical statues and monuments, which required the integration of 3D models. Our initial approach used .obj format for our 3d model files, which we later found out weren't natively supported by ARKit, Apple's framework for creating AR experiences. This discrepancy necessitated a switch to the.usdz format used for 3D models, which is optimized for mobile devices and supported by ARKit[10].

The process of converting.obj samples to.usdz format and, in some cases, obtaining.usdz samples directly presented unexpected challenges to our development workflow. Therefore, we would have to finish all the conversion first to ensure compatibility for our 3D models, so it will be usable in the Arkit environment.

4. **Application crashing when opening AR view** The last important issue is that the application crashes whenever the AR function is disabled. The AR component of our project was designed to overlay detailed 3D models of statues and monuments in the real world environment viewed by the user from the camera. After long debugging and analysis, we determined that the problem stemmed from how AR sessions were managed in the application. In our initial implementation, an AR session instance was reused between invocations of the AR view (when we open an AR view of a new historical location, it's still used the old location session, which might cause problem and crashing). The solution is to modify our session management process to ensure that AR sessions are completely terminated and reset when the users leave the AR view. This change meant that every time an AR service was requested, a new instance of the AR session was created rather than trying to reuse an existing session. This approach greatly improved application performance and stability, and eliminated accidents which previously ruined the AR experience from there. This technical limitation highlighted the importance of careful resource management in mobile applications, especially when dealing with advanced features such as augmented reality.

4 Conclusions

In conclusion, since this was our first project of such type where the scale of the project was massive we faced multiple challenges and it was an arduous journey. We faced many types of problems from technological problems to problems with teammates as well as managing structure. However we as a team believe we have overcame such challenges significantly and have gained experience to avoid doing common mistakes we did at the start of such project. We believe we have gained signification knowledge on the technology we used in such case both Swift UI and React native as well as software practices such as agile methodology and how it works in the real world. An article by the title of "Overview and Guidance on Agile Development in Large Organizations"[3] explains that agile methodology in large organisations is hard to adopt, this was the case for us at the start and even after gaining significant experience we believe that it is extremely hard for us to adopt agile methodology. However we found that such reason does not mean we should not use agile method-

ology, as the article stated that agile methodology is extremely effective in terms of cost of development and we found that such can be the case if agile methodology is adapted properly. This is because team structure and team development can progress faster.

This is also the first time we have worked with real users, and we have felt the challenge of working with real users in real life, away from the pure coursework. We got a good exercise through every communication and exchange with the customer, we now have a certain ability to report to the customer and analyse the current work, these skills will be conducive to our future work in general scenarios.

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